PROJECT TIDEGLASS

# The Chronicle Edition Intelligence and Comparison System

Governing Architecture and Product Requirements Document

| Governing Principle |
| --- |
| Chronicle editions must be compared by meaning, not by raw JSON shape. Every Tideglass comparison must be deterministic, source-bound to two exact immutable edition identities, spoiler-safe for its audience, historically truthful, and human-readable without inventing changes that the source editions do not support. |

Version 1.0 | August 8, 2026

Repository baseline reviewed: Kgray44/treasurehuntSoT @ f1c2f22dd935322c1a71eb80c51592f243dc196d

# Document. Control and Authority

| Field | Value |
|---|---|
| Program | Project Tideglass |
| Subsystem | The Chronicle Edition Intelligence and Comparison System |
| Document type | Governing architecture and product requirements |
| Version | 1.0 |
| Date | August 8, 2026 |
| Status | Governing baseline |
| Repository baseline reviewed | Kgray44/treasurehuntSoT at f1c2f22dd935322c1a71eb80c51592f243dc196d |
| Primary product surfaces | Chronicle detail, edition selection, Personal Journey history, Creator Studio versions, Captain preflight, Community release/update surfaces |
| Core implementation rule | One deterministic semantic comparison authority over immutable edition truth; no raw JSON diff as user-facing truth; no second versioning system. |

## Authority and precedence

This document is authoritative for Tideglass product behavior, semantic comparison rules, user-facing change intelligence, version-pair selection, spoiler-safe comparison projections, and the Tideglass phase boundaries. It does not supersede One Voyage ownership of published editions, Wayfarer ownership of personal history, Harborlight ownership of Community releases, Drydock ownership of authoring schema validation, or the Voyagewright Continuous Development and Mainline Integration Standard.

At implementation time, the current fetched origin/main is the source-level authority for actual paths, models, APIs, and accepted behavior. The repository baseline above records what this governing document reviewed; it must never be used as an instruction to reset newer accepted work.

# Contents.

# 1. Executive Summary

Project Tideglass turns Voyagewright's existing versioning machinery into a first-class human capability. The platform already has immutable published Chronicle editions, checksums, version-pinned live Voyages, Creator Studio publication history, and personal history capable of remembering the exact edition a person played. Tideglass connects those foundations so a human can understand what changed between any two authorized editions without reading raw snapshots or reverse-engineering branch graphs.

The central product experience is a dedicated What Changed? view. A person selects Version A and Version B and receives a structured, human-readable explanation of meaningful differences: story structure, chapters, blocks, choices, endings, side quests, artifacts, locations, media, accessibility, setup requirements, compatibility, and Creator-authored release notes. Technical storage noise must be filtered out.

The signature integration is Compare to what I played. When a Personal Journey record proves that a person completed an older immutable edition, Tideglass preselects that exact edition as the source and the current recommended playable edition as the target. The person can immediately understand what has changed since their own Voyage.

Tideglass is a read-side intelligence system. It does not edit Chronicles, choose the canonical current edition, mutate live Voyages, rewrite history, replace Harborlight update logic, or invent schema semantics. It consumes authoritative immutable inputs and produces deterministic, audience-safe comparison evidence.

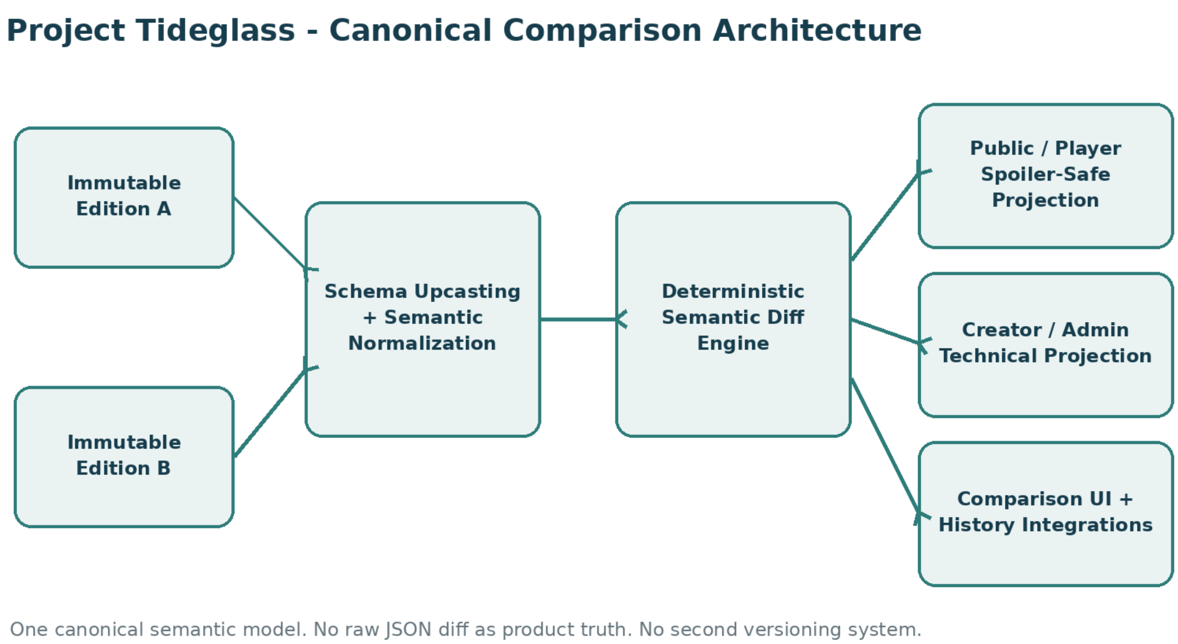


*Figure 1. Tideglass compares two immutable editions through canonical semantic normalization before producing audience-safe change intelligence.*

# 2. Project Identity and Product Vision

## 2.1 Program name

The subsystem is named Project Tideglass. A tideglass is an instrument for looking across changing water and reading where a journey has been and where it now leads. The name fits a system that lets a person look through time at a Chronicle without turning the experience into a code diff wearing a nicer font.

## 2.2 Formal subtitle

The Chronicle Edition Intelligence and Comparison System.

## 2.3 Product vision

Versioning should be visible when it matters and invisible when it does not. Ordinary Players should not need to know internal schema versions or object paths. They should be able to answer simple questions: What changed? Is this version substantially different? Is it worth replaying? Which edition did I experience? What does the Creator recommend now?

## 2.4 Primary use cases

- A Player who completed an older edition wants to know what changed since their Voyage.
- A prospective Player wants to compare two playable editions before joining or replaying.
- A Captain creating a Voyage wants to understand differences between the selected edition and the current recommended edition.
- A Creator wants a precise edition-to-edition summary after publishing.
- Harborlight wants a safe semantic summary of Chronicle changes without exposing raw package internals.
- Support or administrators need a technical comparison view without becoming the owner of version truth.

# 3. Current Platform Context

The current platform already contains the prerequisites that make Tideglass feasible. The generated Feature Catalog records a unified Chronicle platform with checksummed published editions and version-pinned Voyage sessions. Creator Studio already supports immutable publication and version history. Wayfarer personal history preserves exact published-version identity and checksum for a person's historical Voyage. Harborlight retains immutable Community releases and update relationships.

This matters because Tideglass should not build a new version store. Its job is to interpret existing exact versions safely. The authoritative input is the immutable published representation already used by live Voyages and historical records.

| Existing capability | Tideglass use | Ownership remains with |
|---|---|---|
| Checksummed published editions | Source and target comparison identities | One Voyage / publishing |
| Version-pinned live Voyages | Proves active sessions cannot be silently rewritten | One Voyage |
| Personal Chronicle history | Supplies exact edition the person played | Wayfarer / Wakebook |
| Creator publication/version history | Supplies selectable published editions and metadata | Creator Studio / One Voyage |
| Community immutable releases | Provides Community release context and update lineage | Harborlight |
| Drydock schema evolution | Provides canonical upcasting and compatibility rules as it becomes available | Drydock |

| Repository rule |
|---|
| Tideglass may compare authoritative versions. It may not create a parallel definition of what a published Chronicle edition is. |

# 4. Problem Statement

The platform currently possesses more version truth than human-facing version intelligence. A checksum proves that two editions differ, but it does not explain whether the difference is one corrected caption, an entirely new side quest, a reworked finale, a changed arrival provider, or a different Captain requirement.

A raw object diff is not sufficient. Raw snapshots contain serialization choices, ordering details, storage identifiers, schema-shape changes, generated metadata, and other implementation details that may change while the actual Player experience remains semantically identical. Conversely, a tiny raw change may radically alter progression if it rewires a branch or completion condition.

Tideglass therefore needs a semantic comparison layer that understands governed Chronicle concepts and presents only meaningful differences for the requesting audience.

- Prevent raw JSON or ORM diff output from becoming the user experience.
- Prevent schema migrations from creating false change reports.
- Prevent hidden answers, Creator notes, or target-version spoilers from leaking through comparisons.
- Make historical exact-edition records useful to actual people.
- Make version choice understandable before replay or Voyage creation.
- Provide a reusable deterministic contract for later Wakebook, Helm, Shipwright, Harborlight, and Admiralty surfaces.

# 5. Scope and Core Outcomes

Tideglass owns semantic interpretation of differences between two authorized immutable editions of the same canonical Chronicle. It owns comparison records, human-safe categorization, audience-aware redaction, change significance, comparison summaries, and the user-facing comparison experience.

- Exact source and target edition selection.
- Canonical semantic normalization across supported schema versions.
- Deterministic comparison of Chronicle metadata, structure, progression semantics, assets, artifacts, locations, requirements, and accessibility.
- Human-readable change categories and summaries.
- Creator-authored release-note integration without allowing release notes to replace computed truth.
- Spoiler-safe public and Player projections.
- Creator and authorized technical projections.
- Compare-to-what-I-played integration.
- Pre-Voyage edition comparison and selected-version context.
- Compatibility and playability implications.
- Caching and receipts bound to exact checksums and comparison-policy version.
- Accessible, responsive, mobile-safe comparison UI.

# 6. Explicit Non-Goals

- Tideglass does not edit drafts or published editions.
- Tideglass does not publish Chronicles.
- Tideglass does not choose which edition is current, recommended, public, playable, archived, or deprecated. It consumes those policies.
- Tideglass does not mutate Tale Sessions or re-pin active Voyages.
- Tideglass does not rewrite Wayfarer or Wakebook history.
- Tideglass does not replace Harborlight package/update comparison for non-Chronicle Community assets.
- Tideglass does not become Drydock and does not define Story Block validation semantics.
- Tideglass does not compare unrelated Chronicles merely because their JSON looks similar.
- Tideglass does not use fuzzy matching to claim that an unmatched block was renamed or replaced unless an explicit stable relationship exists.
- Tideglass does not expose hidden answers, private variables, Creator-only notes, Captain-only information, invitation secrets, unpublished draft content, or unauthorized private media.
- Tideglass does not use an LLM-generated narrative as authoritative diff truth. Optional prose generation may summarize structured records later, but the structured deterministic diff remains the authority.
- Tideglass does not promise that every historical edition remains playable forever. It reports current playability and compatibility truth supplied by the owning systems.

# 7. Non-Negotiable Design Principles

| Principle | Requirement |
|---|---|
| Semantic truth over storage shape | Compare canonical meaning after schema normalization, not raw JSON serialization. |
| Exact edition identity | Every result is bound to exact source and target edition IDs, checksums, and schema identities. |
| Deterministic comparison | The same source, target, policy version, and audience class produce the same structured result. |
| No invented equivalence | Unmatched entities are added/removed unless an explicit stable identity or replacement relation proves continuity. |
| Spoiler-safe by default | Target-version story details remain hidden unless policy and user action permit reveal. |
| Server-side projection | Unauthorized fields never reach the browser and are not merely hidden with CSS. |
| History is immutable evidence | A person's played-edition anchor comes from Wayfarer/Wakebook truth and is never rewritten to the current edition. |
| No second versioning system | One Voyage/publishing remains the owner of edition identity and playability. |
| Creator notes are annotations | Human release notes supplement but never override detected changes. |
| Explain significance | Major/minor labels must state why; do not fabricate a percentage-different score. |
| Graceful partial comparison | If one subsystem cannot be normalized, report that section as unavailable rather than inventing certainty. |
| Accessible by construction | Change meaning must not rely only on color, animation, side-by-side layout, or icons. |
| Mainline-safe phases | Every Tideglass phase must be independently acceptable into main under the Continuous Development Standard. |

# 8. Cross-Project Ownership and Authority

| System | Tideglass consumes | Tideglass must not own |
|---|---|---|
| One Voyage / Publishing | Published edition identity, immutable snapshots, checksums, playability metadata | Publishing, progression, active-session pinning |
| Wayfarer / Wakebook | Exact historical edition played, historical participation context | History derivation, Memories, Keepsakes, profile privacy |
| Creator Studio / Shipwright | Version lists, Creator annotations, comparison entry points | Draft authoring, block editing, Studio layout authority |
| Drydock | Schema upcasters, strict block semantics, compatibility findings | Validation rules, simulation, publish gate |
| Harborlight | Community release context, update lineage, public listing permissions | Package installation, Community asset diff, remix lineage |
| Helm | Selected Voyage edition and Captain preflight entry point | Captain commands, session recovery, crew operations |
| Admiralty | Privileged diagnostic surface and audit context | Administrative authorization or system ownership |
| Lanternwake | Optional presentation patterns and reduced-motion authority | Comparison truth or semantic diffing |

Tideglass is intentionally a read-side subsystem. Its safest architecture is powerful because it can be deeply integrated without becoming another writer of business state.

# 9. Canonical Domain and Vocabulary

| Term | Definition |
|---|---|
| Edition | An immutable published Chronicle version owned by the canonical publishing domain. |
| Edition Pair | Ordered source and target editions being compared. |
| Semantic Snapshot | Normalized, typed representation of an edition used only for comparison. |
| Comparison Policy Version | Version of Tideglass normalization, classification, and projection rules. |
| Change Record | One deterministic semantic difference with category, kind, evidence, significance, and safe descriptions. |
| Change Set | The complete ordered collection of Change Records for one edition pair. |
| Comparison Summary | Human-oriented aggregation of a Change Set. |
| Audience Projection | Server-generated safe view for public, Player, Creator, Captain, or Administrator contexts. |
| Played Anchor | Exact historical edition proven by a person's personal Voyage record. |
| Recommended Target | Current Creator/platform-recommended playable edition supplied by publishing policy. |
| Compatibility Impact | Whether a change affects platform version, required provider, device, capability, or playability. |
| Creator Annotation | Human-authored release note attached to an edition relationship or edition. |

```
Edition A -> Semantic Snapshot A
Edition B -> Semantic Snapshot B
(A, B, Policy Version) -> Change Set -> Audience Projection -> UI
```

# 10. Edition Identity and Source-of-Truth Rules

Every comparison begins by resolving exact immutable edition identities. Human labels such as “current,” “latest,” “original,” or “the one I played” are selectors, not identities. Before computation, each selector must resolve to one exact published edition ID and checksum.

- The source and target must belong to the same canonical Chronicle for ordinary Tideglass comparison.
- The source may be older or newer than the target. Reverse comparison is permitted but must be labeled clearly.
- The same edition compared to itself returns an explicit NO_MEANINGFUL_CHANGE result, not an empty mysterious page.
- Current and recommended are separate concepts if the publishing domain distinguishes them.
- Historical editions may remain comparable even when no longer playable, provided authorization and retained metadata permit it.
- If a historical edition has been partially redacted for legal/privacy reasons, the comparison must reflect redaction rather than reconstructing removed content.
- Active Tale Sessions remain pinned to their edition regardless of what Tideglass reports about newer versions.

| **No floating aliases in receipts** |
| --- |
| A persisted or cached Tideglass receipt must never say from=current. It must record the exact resolved edition and checksum used at computation time. |

# 11. Canonical Semantic Snapshot Architecture

The comparison engine must not diff the serialized published snapshot directly. Each edition is first transformed into a canonical semantic snapshot. Schema upcasting removes representational differences that do not change meaning; normalization orders sets, resolves known aliases, and converts provider-specific fields into governed semantic forms.

The semantic snapshot is read-only and Tideglass-specific. It is not a new publication format and is not written back into the Chronicle. Its purpose is to create stable comparison input.

| Semantic area | Examples of normalized content |
|---|---|
| Chronicle metadata | title, public description, theme, estimated duration, difficulty, supported crew size, Captain requirements |
| Structure | chapter identity/order, entry/exit points, Story Block identity/order, graph edges |
| Progression semantics | choice targets, conditions, completion modes, timer semantics, provider requirements |
| Artifacts | definition IDs, reveal mode, grant semantics, collection/assembly references |
| World | locations, route references, map bindings, future Landfall provider metadata |
| Media | semantic asset role, checksum/content identity, captions/transcripts, fallback availability |
| Accessibility | required alternatives, reduced-motion/static alternatives, transcript/caption state |
| Safety and requirements | content warnings, physical requirements, helper/provider requirements, minimum platform version |

Storage keys, database row IDs that are not semantic identities, generated derivative filenames, timestamps unrelated to publication meaning, and object serialization order must not produce user-facing changes.

# 12. Entity Identity and Matching Rules

Tideglass must distinguish exact identity from resemblance. The authoritative engine uses stable Chronicle, chapter, block, artifact, location, and relationship identifiers where the publishing model preserves them. When a Creator deletes one block and creates a new one that happens to contain similar text, Tideglass must not silently call it a rename.

| Identity outcome | Meaning | Allowed classification |
|---|---|---|
| EXACT_STABLE_ID | Same governed identity exists in both editions | MODIFIED, MOVED, REWIRED, UNCHANGED |
| EXPLICIT_REPLACEMENT | Publishing/authoring metadata explicitly links replacement | REPLACED with before/after identities |
| UNMATCHED_SOURCE | Source identity absent from target | REMOVED |
| UNMATCHED_TARGET | Target identity absent from source | ADDED |
| AMBIGUOUS | Corrupt or conflicting identity evidence | COMPARISON_SECTION_UNAVAILABLE or diagnostic error |

Heuristic similarity may be used only as non-authoritative Creator assistance, clearly labeled as a suggestion. It must never alter the canonical change record without explicit confirmation that creates a stable replacement relationship owned by the authoring domain.

# 13. Deterministic Difference Engine

1. Resolve and authorize source and target editions.
2. Upcast both editions to supported semantic schemas.
3. Normalize each semantic domain independently.
4. Match entities using exact stable identity or explicit replacement relations.
5. Compare scalar semantics, ordered semantics, set semantics, graph semantics, and provider semantics using type-specific comparators.
6. Emit atomic Change Records.
7. Aggregate related records into human-level groups without losing atomic evidence.
8. Assign category, kind, significance, compatibility impact, and spoiler classification using deterministic rules.
9. Compute a source-bound comparison digest.
10. Project the result for the requesting audience.

The engine must be pure enough to run in unit tests with frozen fixtures and deterministic enough to produce stable evidence for Sounding Line. External AI services are not required for correctness.

```
ComparisonIdentity = SHA256(
  sourceEditionChecksum + targetEditionChecksum +
  semanticSchemaVersion + comparisonPolicyVersion
)
```

# 14. Change Taxonomy

Every atomic change receives one primary semantic category and may receive secondary tags. The taxonomy should be stable enough for filtering and testing, but extensible through versioned registry rules.

| Primary category | Representative changes |
|---|---|
| STORY_CONTENT | Narrative, clue, hint, chapter title, optional narrative content |
| STRUCTURE | Chapter/block added or removed, order changes, entry/exit changes |
| BRANCHING_AND_CHOICES | Choice added/removed, branch target changed, alternate path added |
| ENDING | Ending added/removed/changed, finale requirements changed |
| COMPLETION | Answer/completion mode, Captain approval, timer, provider requirement changed |
| SIDE_QUEST | Optional objective added, removed, changed, or reconnected |
| ARTIFACT | Artifact definition, reveal, grant, collection, assembly behavior changed |
| LOCATION_AND_MAP | Location, route, region, map binding, arrival requirement changed |
| MEDIA | Image, audio, video, model, transcript, caption, poster, fallback changed |
| ACCESSIBILITY | Alternative, transcript, caption, reduced-motion/static support changed |
| SETUP_REQUIREMENTS | Player count, Captain requirement, physical props, travel/setup changed |
| COMPATIBILITY | Minimum platform version, provider/device requirement, deprecated feature changed |
| SAFETY_AND_WARNINGS | Content warning, location safety, privacy-sensitive requirement changed |
| PRESENTATION_METADATA | Cover, public description, theme, estimated duration, difficulty label changed |

# 15. Significance and User-Impact Model

Tideglass must not report a fake “42% changed” score. Different changes have different experiential weight, and a small graph edit can matter more than a large text rewrite. Instead, the system uses an explainable ordinal significance model.

| Level | Meaning | Examples |
|---|---|---|
| PRESENTATION_ONLY | No governed progression or requirement change | cover art, typo, description polish |
| MINOR | Small experience change unlikely to justify replay alone | hint wording, small media swap |
| MEANINGFUL | Noticeable new or changed experience | new clue, changed side quest, new artifact |
| MAJOR | Substantial route, chapter, ending, or requirement change | new chapter, branch restructure, new finale |
| TRANSFORMATIVE | Edition is materially different across several major systems | major rewrite with new structure and outcomes |

Compatibility-critical is a separate flag rather than a significance level. A tiny technical change may be compatibility-critical without making the story feel different. Every summary must list the reasons that produced its level.

# 16. Spoiler, Privacy, and Audience Safety

A comparison can leak future story structure even when neither source edition is secret in the ordinary sense. Tideglass therefore treats comparison output as a projection problem, not merely a formatting problem.

| Audience mode | Default visibility |
|---|---|
| PUBLIC_PREVIEW | Only preview-safe categories and high-level counts; no hidden story details or secret identities. |
| PLAYER_UNPLAYED_TARGET | High-level changes; target-version details hidden behind spoiler reveal where allowed. |
| PLAYER_PLAYED_SOURCE | May reveal safe source-edition details already experienced; target secrets remain hidden by default. |
| PLAYER_PLAYED_BOTH | May reveal more detailed player-safe before/after content for both editions. |
| CREATOR_FULL | Exact authored semantic differences including technical block/graph details, excluding credentials/private account data. |
| ADMIN_DIAGNOSTIC | Creator-level technical comparison plus operational identifiers necessary for support, still subject to private-content authorization. |

- Accepted answers are never exposed merely because text changed.
- Creator-only notes and Captain-only notes never enter Player/public comparison DTOs.
- Private real-world locations remain generalized according to Landfall/Harborlight privacy policy.
- Participant media or private assets respect current consent and legal-retention state.
- A spoiler-reveal control changes presentation of already-authorized data; it does not fetch unauthorized data.

# 17. Creator Release Notes and Annotations

Computed change intelligence and Creator release notes serve different purposes. Tideglass should show both, but never confuse them.

| Layer | Purpose | Authority |
|---|---|---|
| Detected changes | What the editions actually differ on semantically | Tideglass deterministic engine |
| Creator release notes | Why the Creator made the update; editorial context | Creator-authored annotation |
| Compatibility notes | Current platform/provider implications | Owning platform/provider systems projected through Tideglass |
| Historical note | Deprecated or restored edition context | Publishing/Community metadata |

Creator notes may be marked preview-safe, spoiler, technical, or internal. Notes tied to adjacent versions may be shown when comparing that pair. When comparing nonadjacent versions, Tideglass may aggregate notes along the version chain but must compute the actual semantic diff directly from the selected endpoints rather than summing pairwise diffs.

If a Creator note contradicts computed truth, Creator Studio should warn. The public comparison still reports detected truth and labels the Creator note separately.

# 18. Version Selection and Availability Semantics

Tideglass presents version choices, but the publishing domain decides which editions are playable or visible. The comparison UI must clearly distinguish availability from historical existence.

| Badge / state | Meaning |
|---|---|
| CURRENT_RECOMMENDED | Edition currently recommended by the Creator/platform for new Voyages. |
| PLAYED_BY_YOU | Exact edition appears in the signed-in person's history. |
| ORIGINAL | Earliest retained published edition. |
| PLAYABLE | May currently be selected for a new Voyage under owning policy. |
| HISTORICAL_ONLY | Retained for history/comparison but not selectable for new Voyages. |
| DEPRECATED | Superseded and discouraged; reason may be shown. |
| INCOMPATIBLE | Cannot run on current platform/provider conditions. |
| REDACTED | Historical metadata retained but some content cannot be displayed. |

The UI should never imply that “latest” equals “best,” “recommended,” or “the edition you played.” Those are separate facts.

# 19. Compare to What I Played

This is the signature Player feature. A signed-in person viewing an eligible Chronicle or their Journey Archive should be able to choose Compare to what I played. Tideglass resolves the exact edition from Wayfarer/Wakebook, then compares it with the current recommended target edition.

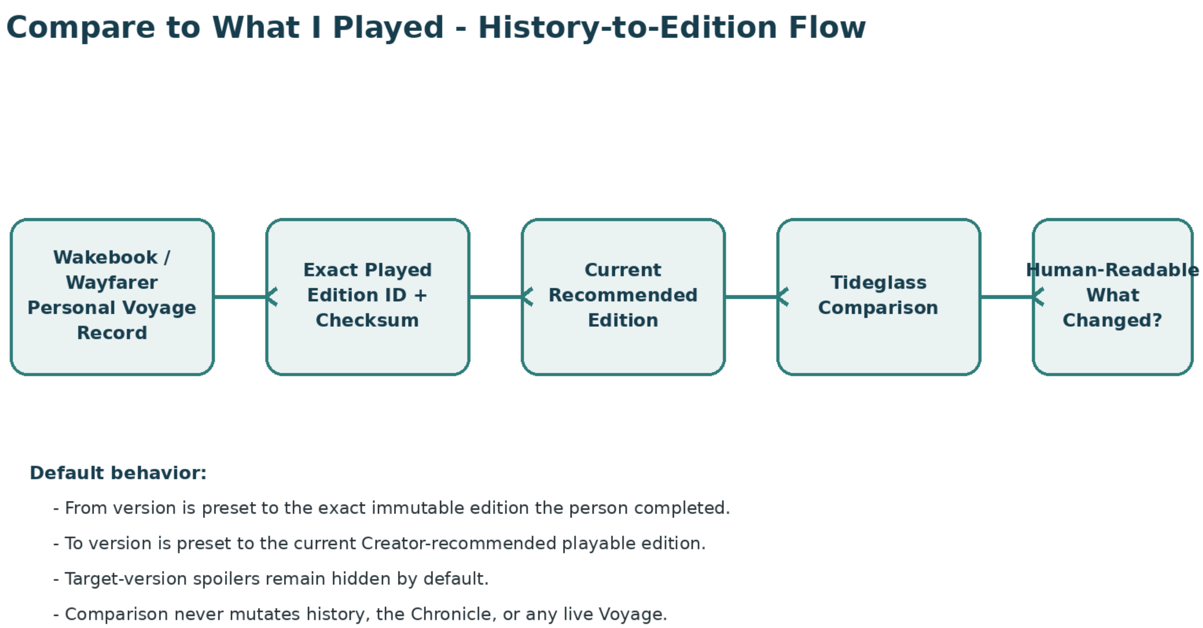


*Figure 2. Compare to What I Played uses personal historical truth only to select the source edition; Tideglass never rewrites that history.*

- If the person completed the Chronicle multiple times, show a compact played-edition chooser with dates and outcomes.
- If the person already played the current recommended edition, show You are up to date and permit comparison with other editions manually.
- If the source edition is historical-only, comparison remains available when authorized.
- If the target is unavailable, Tideglass may compare to the most recent authorized edition but must explain the substitution.
- The entry action should be available from Wakebook history and the Chronicle detail page, not hidden behind manual URL editing.

# 20. Pre-Voyage and Chronicle Detail Comparison

Tideglass should make version choice understandable before a replay or Voyage is created. A Chronicle detail page may expose a Version section containing the recommended edition, other playable editions, the person's previously played editions, and a Compare action.

The default comparison page should emphasize experiential differences rather than technical internals. A typical top section includes source and target edition cards, publication dates, status badges, an overall significance label, a one-paragraph safe summary, and category counts.

- Version A and Version B selectors with labels rather than raw IDs.
- Swap direction control.
- Compare to newest/recommended shortcut.
- Compare to what I played shortcut when history exists.
- Filter chips for Story, Structure, Artifacts, Locations, Media, Accessibility, Requirements, Compatibility.
- Spoiler control with safe default.
- Before/after details only where authorized and useful.
- Clear action to select the chosen playable edition for the owning workflow; Tideglass itself does not perform the selection mutation.

# 21. Wayfarer and Wakebook Integration

Wayfarer owns durable personal historical truth. Wakebook will own the rich Personal Journey Archive experience. Tideglass consumes only the minimum history projection needed to identify an exact source edition and to label personally meaningful context.

| History fact | Tideglass use | Restriction |
|---|---|---|
| Published version ID/checksum | Select exact source edition | Never substitute current edition |
| Completion date | Label played context | No public exposure without history policy |
| Crew/ending summary | Optional context around a played anchor | Not used to infer version differences |
| Multiple playthroughs | Offer source chooser | Do not merge separate history records |
| Memory/Keepsake | Optional link back to Journey detail | Tideglass does not read private content for diffing |

A Tideglass result may link back to the historical Voyage detail. It must never copy Memories, reflections, participant consent, or other private Wayfarer content into comparison caches.

# 22. Creator Studio and Shipwright Integration

Creator Studio should use Tideglass to help Creators understand published evolution. Shipwright may later provide richer comparison presentation inside Studio, but Tideglass remains the comparison service.

- Versions list gains Compare actions.
- After publishing, Studio may show a detected change summary against the previous recommended edition.
- Creator release-note authoring may sit alongside the detected summary.
- Technical Creator mode may reveal block IDs, graph edges, normalized fields, and exact before/after values where safe.
- Draft-to-published comparison is a future adapter and must not be confused with authoritative edition-to-edition comparison unless the draft is converted through the same semantic normalization contract.
- Shipwright must not implement a second diff engine in the frontend.

| Drydock boundary |
| --- |
| Drydock defines strict authoring semantics and schema evolution. Tideglass consumes those semantics. Studio presents them. Nobody gets three competing interpretations of what a Choice block means. |

# 23. Harborlight and Community Integration

Harborlight already owns Community Listing and immutable Community Release update behavior. Tideglass may provide Chronicle-specific semantic change summaries when a Community release points to a changed published Chronicle edition.

- Harborlight remains authoritative for package files, dependencies, licensing, installation, rollback, remix lineage, and Community release visibility.
- Tideglass may provide a What changed in the Chronicle? semantic panel alongside Harborlight's package/update summary.
- Active Tale Sessions remain pinned and cannot be mutated by Community updates.
- Public Community comparisons use the PUBLIC_PREVIEW Tideglass projection.
- Remixed or forked Chronicles are not automatically treated as two editions of one Chronicle; explicit lineage rules are required before any cross-lineage comparison is presented as authoritative.

# 24. Helm and Captain Integration

Helm can use Tideglass during Voyage creation and preflight. A Captain selecting an older or nonrecommended playable edition should be able to inspect meaningful differences from the recommended version before launch.

- Show selected edition versus recommended edition.
- Highlight changed Captain requirements, crew size, providers, location requirements, physical props, accessibility requirements, and estimated duration.
- Do not expose Player-hidden story answers merely because the requester has Captain authority. Operational projection and Creator-full projection are distinct.
- Once a Tale Session launches, Tideglass becomes informational only. It must never change the session's pinned edition.

# 25. Drydock and Schema-Evolution Integration

Drydock is the preferred owner of strict Story Block contracts, semantic schema evolution, and historical upcasters. Tideglass must integrate with those contracts rather than re-implementing them.

When both editions can be upcast to a common semantic schema, the comparison proceeds normally. When one subsystem cannot be upcast safely, Tideglass should compare unaffected sections and mark the incompatible section explicitly unavailable.

| Compatibility case | Required Tideglass behavior |
| --- | --- |
| Both versions current schema | Normal semantic comparison |
| Older version upcasts losslessly | Compare normalized forms; do not report migration-only changes |
| Older version upcasts with declared loss | Mark affected details partial and explain limitation |
| Unsupported historical schema | Fail affected section closed; retain edition-level metadata comparison |
| Provider contract unknown | Report compatibility uncertainty without inventing semantics |

# 26. Data Model and Service Contracts

Tideglass should remain primarily computed/read-side. Persistent data should be limited to durable Creator annotations, optional comparison caches, and evidence receipts where operational value justifies them.

```
type EditionPair = {
  chronicleId: string;
  sourceEditionId: string;
  sourceChecksum: string;
  targetEditionId: string;
  targetChecksum: string;
};

type ChronicleChangeRecord = {
  id: string;
  category: ChronicleChangeCategory;
  kind: "ADDED" | "REMOVED" | "MODIFIED" | "MOVED" | "REWIRED" | "REPLACED";
  entityType: string;
  entityId?: string;
  significance: ChangeSignificance;
  spoilerLevel: ComparisonSpoilerLevel;
  compatibilityImpact?: CompatibilityImpact;
  evidence: ChangeEvidence;
};
```

| Concept | Persistence recommendation |
|---|---|
| Semantic Snapshot | Computed; cacheable by edition checksum + semantic schema version |
| Change Set | Computed; optional cache by exact pair + policy version |
| Comparison Receipt | Persist if needed for diagnostics/evidence; immutable identity |
| Creator Annotation | Persisted, version-bound, authored and audited |
| Audience Projection | Computed per request; do not persist private Player context |
| Played Anchor | Never duplicated; resolved from Wayfarer/Wakebook |

# 27. API and Projection Contracts

```
GET /api/chronicles/:chronicleId/editions
GET /api/chronicles/:chronicleId/comparison?from=<edition>&to=<edition>&mode=<safe-mode>
GET /api/chronicles/:chronicleId/comparison/played-anchor
POST /api/studio/chronicles/:chronicleId/editions/:editionId/annotations
```

Exact routes may follow current repository conventions. The important requirement is typed server-side authorization and projection. Clients must not request a Creator-full DTO by sending mode=creator and trusting the server to admire their initiative.

| DTO | Audience | Key restrictions |
|---|---|---|
| EditionSummaryDTO | Authorized users | No private draft or secret storage metadata |
| PublicComparisonDTO | Public | Preview-safe only |
| PlayerComparisonDTO | Signed-in Player | History-aware; target spoilers hidden by policy |
| CreatorComparisonDTO | Authorized Creator | Technical semantic details, no account/private operational secrets |
| AdminComparisonDTO | Privileged support | Technical diagnostics plus governed support context only |

# 28. Product UI and Interaction Architecture

The dedicated comparison experience should feel like a Chronicle history instrument, not a Git diff viewer. On desktop, use a broad readable change feed with a compact version header and optional category rail. On mobile, stack source/target selectors and preserve readable cards without horizontal scrolling.

- Top summary: source edition, target edition, status badges, dates, overall significance, safe summary.
- Change overview cards: counts and key reasons by category.
- Detailed feed grouped by category and Chronicle structure.
- Optional side-by-side details only for short, safe text or metadata.
- Creator technical mode can expose IDs and exact field changes in collapsible diagnostics.
- The same change has one accessible textual statement even when icons, color, or motion reinforce it.
- No empty comparison page: if there are no changes, explicitly say the editions are semantically equivalent under the current policy.
- Every screen includes a clear route back to the Chronicle, Journey Archive, Studio, or Captain flow that opened it.

# 29. Accessibility and Universal Language

- Comparison meaning must remain complete at 200% zoom and narrow mobile widths.
- Do not require horizontal side-by-side reading to understand a change.
- Use headings and landmarks so screen-reader users can navigate categories and change groups.
- Added/removed/changed state must be stated in text, not only green/red color.
- Spoiler reveal controls require clear labels, focus management, and announcements.
- Reduced motion must preserve all comparison information; animation is decorative/supportive only.
- Before/after media must provide accessible names, captions, transcripts, or governed fallbacks.
- Technical Creator details should use semantic tables/lists rather than inaccessible raw formatted JSON dumps.
- Universal Language terminology must use current Voyagewright terms: Chronicle, edition, Voyage, Player, Captain, Creator, Journey Archive, and Community Harbor as governed.

# 30. Performance, Caching, and Determinism

Because editions are immutable, Tideglass is unusually cache-friendly. A comparison result can be keyed by the exact pair of checksums, semantic schema version, and comparison-policy version. Public/Creator projections may be cached separately from account-specific Player context.

- Never invalidate a comparison because a mutable draft changed. Only source/target edition identity or comparison policy changes matter.
- Cache canonical Change Sets, not unsafe user-specific projections.
- Do not cache private played-anchor context into a shared result.
- Large Chronicles should support streamed or paginated detailed change groups after a compact summary is available.
- Performance tests should include Chronicles with large graphs, many assets, many artifacts, and long version histories.
- The engine should avoid O(N^2) fuzzy matching because authoritative matching uses stable identity.

Recommended target: common edition comparisons should produce a summary in well under one second on server hardware after normalization cache warmup; uncached large Chronicles should remain bounded and observable. Exact budgets should be finalized during implementation against the repository's real data sizes.

# 31. Failure, Fallback, and Observability

| Failure | User behavior | Diagnostic behavior |
|---|---|---|
| Edition not authorized | Clear access-denied state | Audit safe identifier, no content leak |
| Edition missing/corrupt | Comparison unavailable with support path | Correlation ID and exact internal cause |
| Schema unsupported | Partial or unavailable section | Record schema versions and missing upcaster |
| Asset unavailable | Describe media change without preview | Record asset identity/status, not private storage key |
| Comparison timeout | Retryable summary state | Timing metrics and bounded cancellation |
| Creator annotation unavailable | Detected changes still render | Annotation failure does not invalidate diff |
| History anchor unavailable | Manual version selection remains | Do not guess what the person played |

Operational logs must contain comparison identity, source/target IDs, checksums, policy version, timing, category counts, and failure codes. They must not contain hidden narrative payloads, answers, private media URLs, or account secrets.

# 32. Security and Privacy

- Authorize both source and target editions independently before comparison.
- Derive audience class from the authenticated account and resource relationship server-side.
- Treat source/target IDs as untrusted input and prevent cross-Chronicle IDOR.
- Never expose raw published snapshots to a client solely to compute a diff.
- Prevent public comparison from enumerating private or unlisted historical editions.
- Use CSRF protection for annotation mutations.
- Sanitize Creator annotation content and enforce spoiler classification.
- Do not store a person's history record inside shared comparison caches.
- Redact storage keys, package paths, private location precision, and protected content.
- Rate limit expensive comparison endpoints and Creator annotation mutations.
- Audit privileged diagnostic comparisons when they reveal more than ordinary Player/public projections.

# 33. Testing and Acceptance Matrix

| Area | Required proof |
|---|---|
| Determinism | Same fixtures and policy version produce byte-stable structured change results. |
| Schema normalization | Equivalent semantics across schema versions produce no false changes. |
| Identity matching | Stable IDs modify; deleted/recreated entities become remove/add; no fuzzy false rename. |
| Graph semantics | Edge, branch, ending, condition, and completion changes are detected correctly. |
| Spoiler safety | Public and Player DTOs exclude hidden answers, notes, secrets, and target spoilers. |
| History integration | Played anchor resolves exact edition ID/checksum; multiple playthrough selection works. |
| Authorization | Foreign/private/unlisted edition access fails closed; Creator/admin projections require authority. |
| UI | Desktop/mobile comparison, filters, swap, no-change state, error states, navigation reachability. |
| Accessibility | Keyboard, focus, screen reader semantics, 200% zoom, reduced motion, Axe serious/critical zero. |
| Performance | Large version pairs remain within budgets; cache isolation proven. |
| Cross-project | Wayfarer/Wakebook, Studio, Harborlight, Helm entry points use the same Tideglass service. |
| Mainline safety | Every phase passes its declared Sounding Line gate before merge. |

Acceptance must include examples with meaningful progression differences, not only metadata changes. A test suite that proves two covers differ while never changing a branch is technically green and spiritually unemployed.

# 34. Implementation Phases

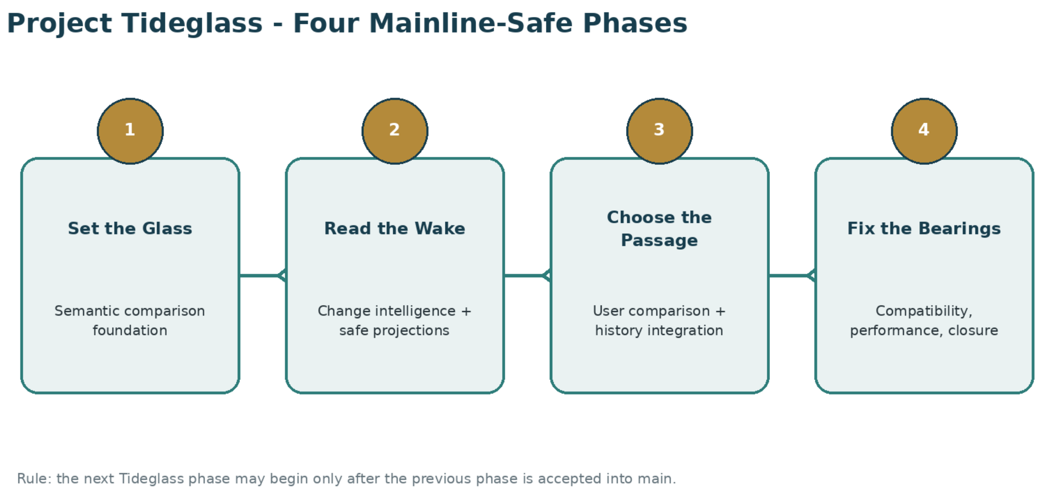


*Figure 3. Tideglass phases are independently mainline-safe; the next Tideglass phase begins from accepted main after the previous phase lands.*

## Phase 1 - Set the Glass

Semantic Edition Difference Foundation. Build exact edition-pair resolution, semantic snapshot contracts, schema-normalization adapters, stable identity matching, atomic Change Records, deterministic diff engine, fixture corpus, diagnostic CLI/service seam, and comparison receipts. User-facing comparison may remain dormant or development-only.

Gate: representative current editions and historical fixtures compare deterministically; existing publication and runtime behavior are unchanged; no new ordinary user surface is required.

## Phase 2 - Read the Wake

Change Intelligence, Significance, Spoiler Safety, Compatibility, and Release Notes. Add full change taxonomy, significance rules, compatibility implications, audience projections, Creator annotations, safe summaries, caching, and API contracts.

Gate: public/Player/Creator projections are privacy-safe, deterministic, and independently useful even without the final polished comparison UI.

## Phase 3 - Choose the Passage

User-Facing Comparison and History Integration. Deliver the polished What Changed? experience, version selectors, compare-to-recommended, Compare to what I played, Chronicle-detail entry points, Wakebook integration, Creator Studio version actions, and Captain preflight consumption where contracts are stable.

Gate: a normal user can discover, navigate, understand, and leave the comparison experience without manual URLs; major flows pass product-governance walkthrough requirements.

## Phase 4 - Fix the Bearings

Cross-Project Completion, Historical Compatibility, Performance, Accessibility, Observability, and Closure. Finalize Drydock upcasting integration, Harborlight and Helm seams, long-history performance, localization/universal language, admin diagnostics, full browser/accessibility matrix, documentation, Feature Catalog integration, and program closure.

Gate: all supported edition histories compare truthfully, all integration surfaces use the canonical service, Sounding Line accepts the project, and the owner-facing product walkthrough passes.

# 35. Mainline Safety Contracts

All Tideglass phases are governed by the Voyagewright Continuous Development and Mainline Integration Standard. Parallel projects are allowed; Tideglass phase stacking is not allowed by default.

| Phase | Mainline-safe state if all later phases are cancelled |
|---|---|
| Phase 1 | A deterministic semantic diff service exists behind internal/development seams. No existing user behavior is broken or forced to use it. |
| Phase 2 | Safe comparison APIs and structured summaries exist and may be consumed by authorized internal surfaces. The absence of Phase 3 leaves no broken public route. |
| Phase 3 | Complete ordinary user comparison experience exists. Phase 4 adds hardening/integrations but is not required for the delivered comparison journey to function. |
| Phase 4 | Project closure with full cross-project integration and final evidence. |

- Each phase branches from current accepted origin/main.
- No next Tideglass phase begins until the prior Tideglass phase is accepted into main.
- Every phase declares Prisma/migration impact before coding; Phase 1 should prefer no business-schema changes unless required for notes/cache evidence.
- If another project advances main, Tideglass reconciles semantically before acceptance and reruns only evidence invalidated under Sounding Line.
- Completed phases converge promptly; they do not sit while later Tideglass phases stack on top.

# 36. Final Acceptance Criteria

- Any two authorized supported editions of the same Chronicle can be compared by exact identity.
- Comparison is semantic and schema-normalized rather than raw serialized-object diffing.
- Every Change Record is traceable to source and target evidence.
- No unsupported fuzzy identity claim can silently become authoritative.
- Public and Player comparisons are spoiler-safe and privacy-safe by server-side projection.
- Creator release notes are clearly distinguished from detected changes.
- Compare to what I played uses exact Wayfarer/Wakebook edition history and works for multiple playthroughs.
- Chronicle detail and history surfaces provide discoverable entry points; no ordinary comparison feature requires manual URL entry.
- Version availability, playability, recommended status, historical-only status, and compatibility are represented truthfully.
- Active Voyages remain pinned and unmodified.
- Harborlight, Helm, Shipwright, Wakebook, Admiralty, and Drydock integrations respect ownership boundaries.
- Comparison caching is source-bound and cannot leak Player-specific context.
- Desktop, mobile, keyboard, reduced-motion, screen-reader, zoom, failure, empty, and no-change states are complete.
- Sounding Line produces the required mainline/release evidence.
- The Feature Catalog records Tideglass only after accepted mainline capability exists.
- The project reaches PRODUCT ACCEPTED only after the actual running comparison experience passes the owner walkthrough where required.

| Final governing rule |
| --- |
| Tideglass may explain how a Chronicle changed. It may never change the Chronicle, rewrite the past, or reveal more of the future than the requesting person is allowed to know. |

# Appendix A. Detailed Change Category Catalog

| Category | Atomic examples | Default public handling |
|---|---|---|
| STORY_CONTENT | text changed, clue added, hint removed | High-level; exact target text spoiler-gated |
| STRUCTURE | chapter/block add/remove/reorder | Counts and safe titles where public |
| BRANCHING_AND_CHOICES | choice count, target edge, conditional route | High-level branch impact by default |
| ENDING | ending count/requirements/result metadata | Spoiler-safe existence only unless revealed |
| COMPLETION | answer mode, approval, timer, provider | Show mechanics without secret answer |
| SIDE_QUEST | objective add/remove/reconnect | Safe title/summary per content policy |
| ARTIFACT | definition/reveal/grant/assembly change | Safe artifact names only if public/revealed |
| LOCATION_AND_MAP | location/route/radius/provider change | Generalize sensitive coordinates |
| MEDIA | asset content identity/fallback/transcript changed | Safe preview if authorized |
| ACCESSIBILITY | caption/transcript/static fallback added/removed | Generally safe and prominently surfaced |
| SETUP_REQUIREMENTS | crew/Captain/physical/device requirement changed | Safe and prominently surfaced |
| COMPATIBILITY | platform/provider/minimum version changed | Safe and prominently surfaced |
| SAFETY_AND_WARNINGS | content warning/privacy/location safety changed | Safe but respect warning-spoiler policy |
| PRESENTATION_METADATA | cover/theme/difficulty/duration/public description | Public if source metadata is public |

# Appendix B. Example Comparison Records


```
{
  "category": "STRUCTURE",
  "kind": "ADDED",
  "entityType": "chapter",
  "entityId": "chapter-observatory",
  "significance": "MAJOR",
  "spoilerLevel": "PREVIEW_SAFE",
  "summary": "A new chapter was added between Chapters 3 and 4.",
  "evidence": {
    "sourceEditionChecksum": "...",
    "targetEditionChecksum": "..."
  }
}

{
  "category": "COMPLETION",
  "kind": "MODIFIED",
  "entityType": "storyBlock",
  "entityId": "block-47",
  "significance": "MEANINGFUL",
  "spoilerLevel": "STORY_SPOILER",
  "publicSummary": "A completion requirement changed.",
  "creatorSummary": "Completion changed from Captain approval to a location provider."
}
```


The public projection may omit entity IDs and technical details. The Creator projection may include them. Both derive from the same canonical Change Record.

# Appendix C. Phase Acceptance Checklist

| Gate | Required proof |
| --- | --- |
| Preflight | Fresh origin/main, clean task worktree, active-phase registration, dependency review |
| Mainline Safety | Phase contract states active/dormant behavior and permanent-stop safety |
| Schema | Migration reservation declared; SQLite/MySQL parity if schema changes |
| Focused validation | Semantic engine, security, privacy, integration tests for changed contracts |
| Reconciliation | Current origin/main reviewed immediately before acceptance |
| Sounding Line | Required authoritative gate and evidence receipt |
| Product reality | For user-facing phases: navigation, desktop/mobile, empty/error/no-change, owner walkthrough gate |
| Merge | Phase integrated into main with post-merge proof and remote parity |
| Cleanup | Task resources and worktree handled; no stale generated output |
| Next phase | Starts only from newly accepted main |

# Glossary and References.

| Reference | Use in this document |
| --- | --- |
| Current GitHub main @ f1c2f22dd935322c1a71eb80c51592f243dc196d | Repository implementation baseline reviewed for project creation. |
| Development_Docs/Features/FEATURE_CATALOG.md | Confirms mainline immutable Chronicle platform, checksummed editions, version-pinned Voyages, account and workspace foundations. |
| Project Wayfarer governing and personal-history records | Exact version-pinned personal Voyage history, historical edition identity, privacy projections. |
| Project Harborlight governing and update architecture | Immutable Community releases, update/version relationships, active-session pinning. |
| Project Drydock Governing Document | Strict authoring semantics, schema evolution, historical compatibility, simulation and publishing verification. |
| Voyagewright Global Product Governance Standard | User-facing discoverability, journey quality, visual/product acceptance requirements. |
| Voyagewright Continuous Development and Mainline Integration Standard v1.0 | Phase-level convergence, Mainline Safety Contracts, concurrency and migration governance. |
| Project Feature Catalog Review / Tideglass concept record | Original product intent: semantic edition comparison, Compare to what I played, pre-Voyage selection, human-readable change intelligence. |

## Glossary

- Chronicle: canonical authored experience.
- Edition: immutable published Chronicle version.
- Voyage: live version-pinned playthrough/session.
- Played Anchor: exact edition proven by personal Voyage history.
- Semantic Snapshot: normalized comparison-only representation of an edition.
- Change Set: deterministic collection of semantic changes for an edition pair.
- Audience Projection: authorized safe DTO derived from a Change Set.
- Recommended Edition: current edition preferred for new play under owning policy; not necessarily newest by timestamp.
- Historical-Only Edition: retained for history/comparison but not currently selectable for new Voyages.

| Document status |
| --- |
| Version 1.0 is the governing project baseline. Implementation prompts must revalidate current origin/main, preserve newer accepted architecture, and follow the permanent Continuous Development and Mainline Integration Standard. |